

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	03 October 2026
Team ID	6ab7f10630f9767f87728ca3
Project Name	Import Data Using Transform Maps (Spreadsheet)
Team Leader	E Dilli Kumar
Team Members	R Vinish, K Jerina, B Mohanavalli, P Sri Ragavan
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration & Login	Registration through Form Registration through Gmail Registration through LinkedIn Role-based login (Admin / Data Manager)
FR-2	User Confirmation	Confirmation via Email Confirmation via OTP
FR-3	Spreadsheet Upload (Import Source)	Upload spreadsheet files (.xlsx, .xls, .csv) Create an Import Set / staging table from the uploaded file Validate file format, size and column headers
FR-4	Transform Map Creation & Configuration	Create a Transform Map between the staging (source) table and the target table Field mapping (source column to target field) Coalesce on key fields to prevent duplicate records Field map scripts / transform scripts for data conversion
FR-5	Data Transformation & Import	Run the transform to load data into the target table Insert new records and update existing records Handle null values, data type conversion and reference fields
FR-6	Error Handling & Logging	Display import status and transform history Report skipped / failed rows with the reason for failure Download an error log report
FR-7	Data Verification & Reporting	Preview imported data in the target table Summary report: rows inserted, updated, ignored and failed

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	Simple, intuitive interface for uploading spreadsheets and mapping fields, with clear guidance and error messages so non-technical users can import data without coding.
NFR-2	Security	Role-based access control, secure authentication (OTP / OAuth), encrypted data transfer (HTTPS) and validation of uploaded files to prevent malicious content and unauthorized access.
NFR-3	Reliability	Imports are consistent and accurate; failed rows are logged without corrupting existing data, and the import can be safely re-run (coalesce prevents duplicates).
NFR-4	Performance	Large spreadsheets are processed in reasonable time through batch/bulk processing, with fast field mapping and transformation and minimal impact on system response.
NFR-5	Availability	The import and transform functionality is available at least 99% of the time, with scheduled maintenance windows and data backup and recovery.
NFR-6	Scalability	The solution handles growing file sizes, record counts and numbers of concurrent users and supports adding new transform maps and target tables without redesign.